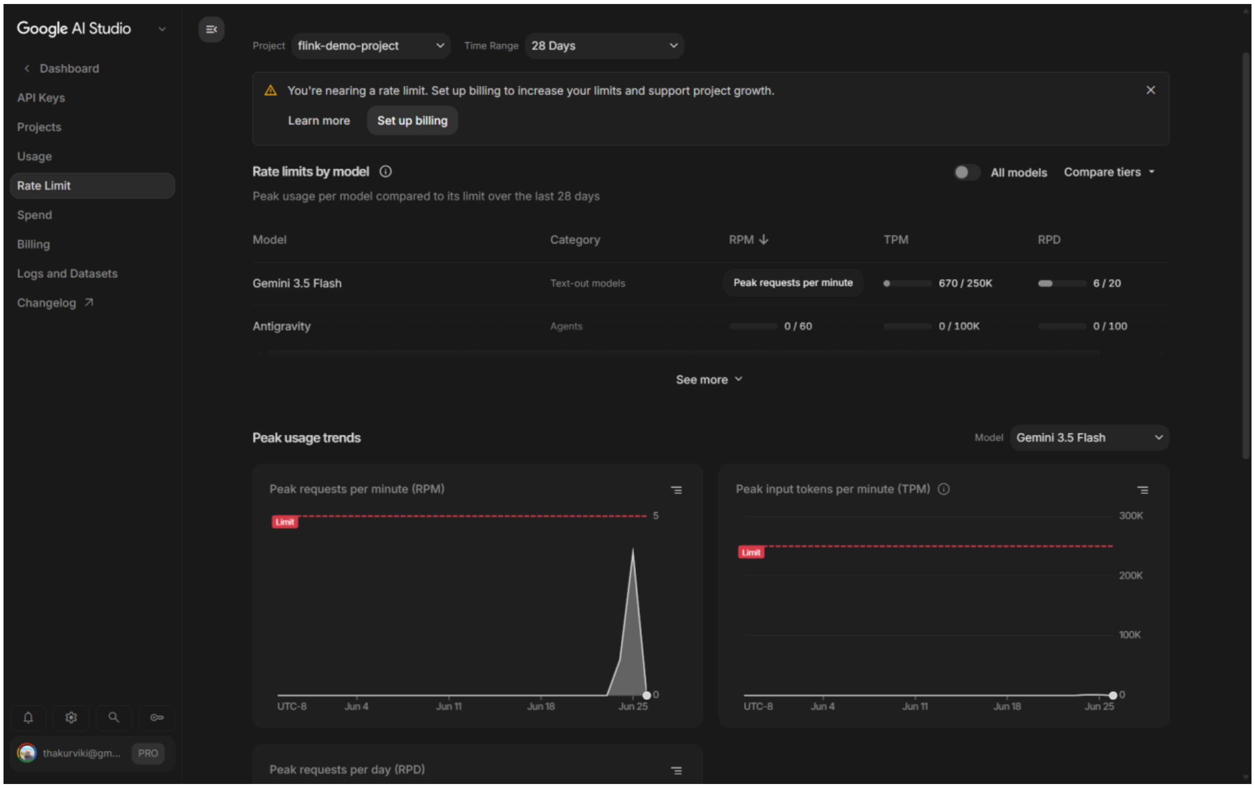
Usage of Google Gemini AI Studio -



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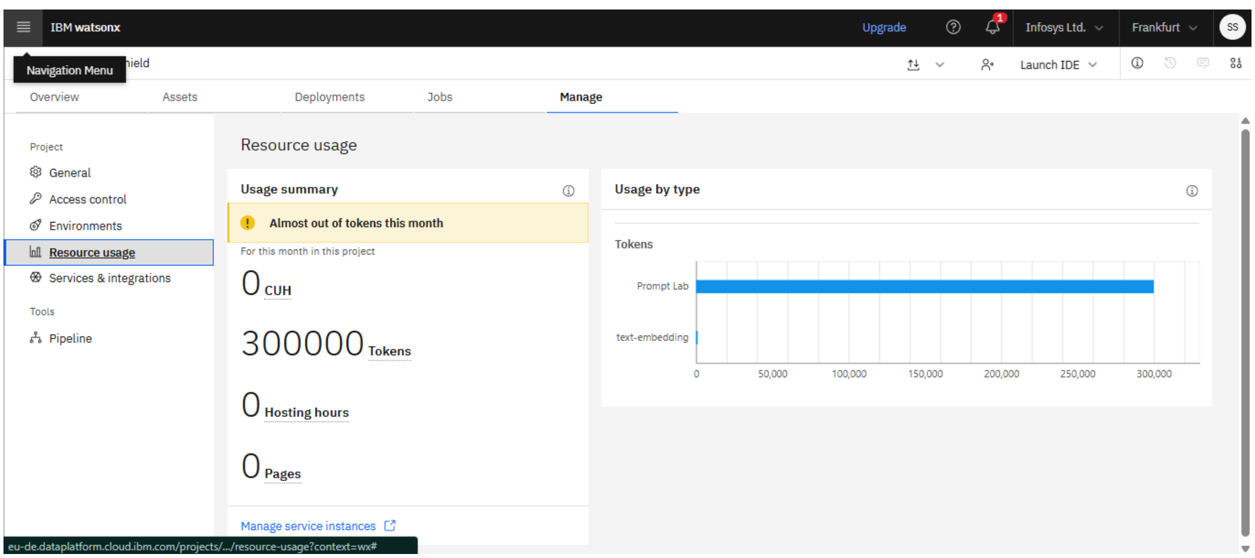
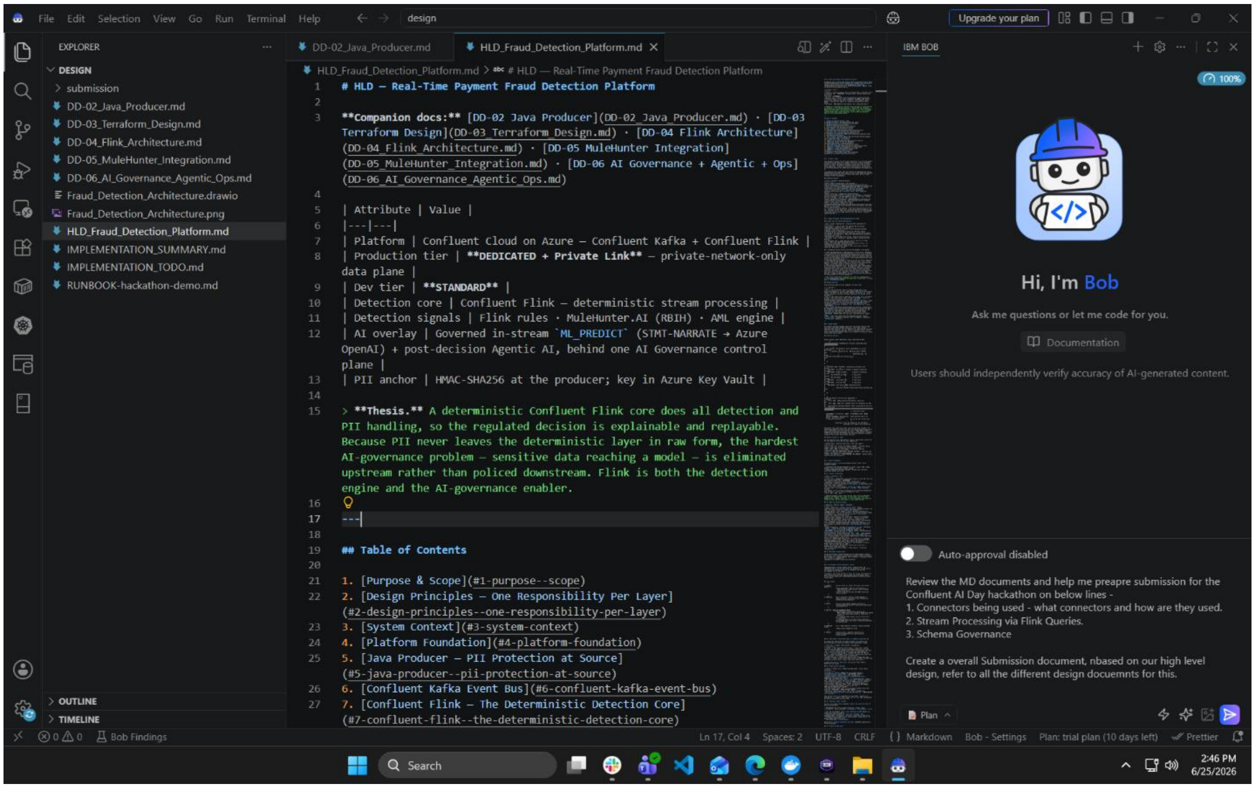


IBM Bob Usage in the project –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DD-03_Terraform_Design.md
DD-04_Flink_Architecture.md
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DD-06_AI_Governance_Agentic_Ops.md
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HLD_Fraud_Detection_Platform.md
HLD_Fraud_Detection_Platform.md > # HLD — Real-Time Payment Fraud Detection Platform
HLD — Real-Time Payment Fraud Detection Platform
Companion docs: [DD-02 Java Producer](DD-02_Java_Producer.md) · [DD-03 Terraform Design](DD-03_Terraform_Design.md) · [DD-04 Flink Architecture](DD-04_Flink_Architecture.md) · [DD-05 MuleHunter Integration](DD-05_MuleHunter_Integration.md) · [DD-06 AI Governance + Agentic + Ops](DD-06_AI_Governance_Agentic_Ops.md)
Attribute	Value
Platform	Confluent Cloud on Azure — Confluent Kafka + Confluent Flink
Production tier	**DEDICATED + Private Link** — private-network-only data plane
Dev tier	**STANDARD**
Detection core	Confluent Flink — deterministic stream processing
Detection signals	Flink rules · MuleHunter.AI (RBIH) · AML engine
AI overlay	Governed in-stream `ML_PREDICT` (STMT-NARRATE → Azure OpenAI) + post-decision Agentic AI, behind one AI Governance control plane
PII anchor	HMAC-SHA256 at the producer; key in Azure Key Vault
> **Thesis.** A deterministic Confluent Flink core does all detection and PII handling, so the regulated decision is explainable and replayable. Because PII never leaves the deterministic layer in raw form, the hardest AI-governance problem — sensitive data reaching a model — is eliminated upstream rather than policed downstream. Flink is both the detection engine and the AI-governance enabler.

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3. [System Context](#3-system-context)
4. [Platform Foundation](#4-platform-foundation)
5. [Java Producer — PII Protection at Source](#5-java-producer--pii-protection-at-source)
6. [Confluent Kafka Event Bus](#6-confluent-kafka-event-bus)
7. [Confluent Flink — The Deterministic Detection Core](#7-confluent-flink--the-deterministic-detection-core)
IBM BOB
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Hi, I'm Bob
Ask me questions or let me code for you.
Documentation
Users should independently verify accuracy of AI-generated content.
Auto-approval disabled
Review the MD documents and help me preapre submission for the Confluent AI Day hackathon on below lines -
1. Connectors being used - what connectors and how are they used.
2. Stream Processing via Flink Queries.
3. Schema Governance
Create a overall Submission document, nbased on our high level design, refer to all the different design documnts for this.
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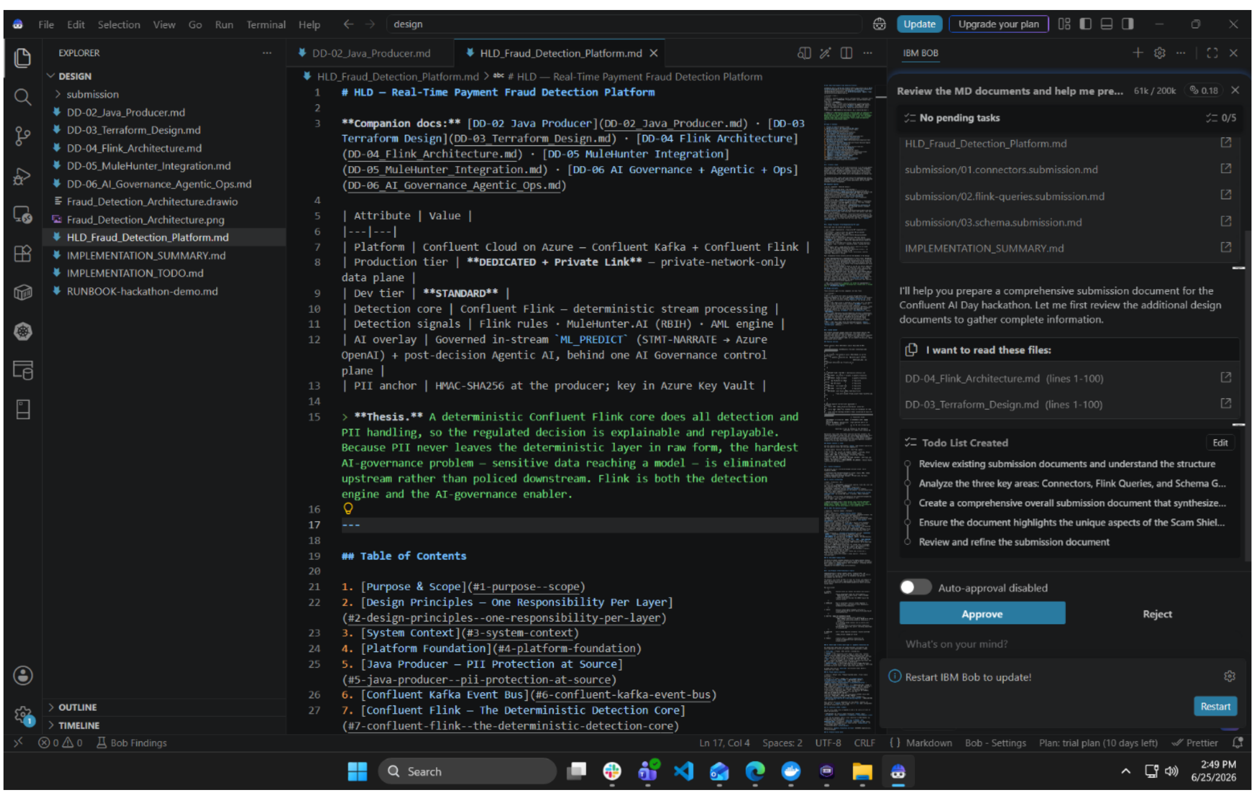
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submission/03.schema.submission.md
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I'll help you prepare a comprehensive submission document for the Confluent AI Day hackathon. Let me first review the additional design documents to gather complete information.
I want to read these files:
DD-04_Flink_Architecture.md (lines 1-100)
DD-03_Terraform_Design.md (lines 1-100)
Todo List Created
Edit
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Create a comprehensive overall submission document that synthesize...
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What's on your mind?
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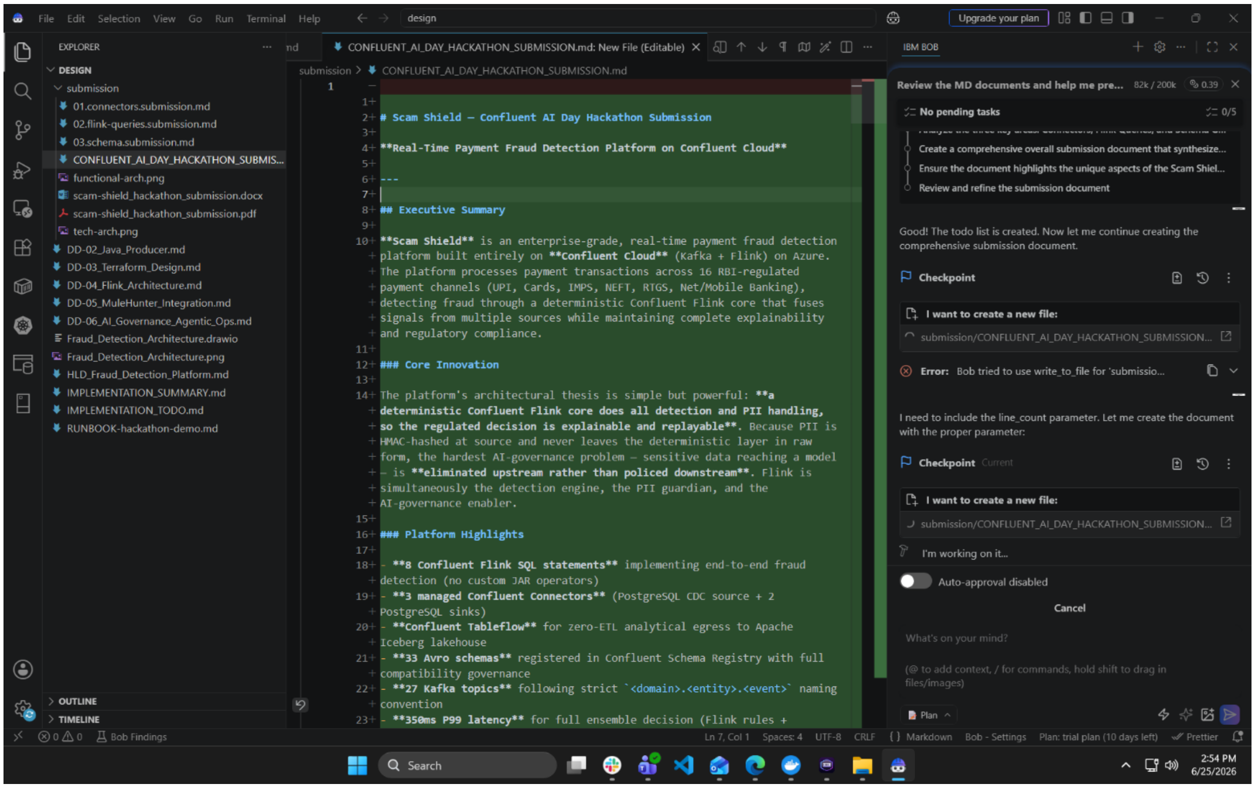

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Scam Shield — Confluent AI Day Hackathon Submission
Real-Time Payment Fraud Detection Platform on Confluent Cloud
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Executive Summary
Scam Shield is an enterprise-grade, real-time payment fraud detection platform built entirely on **Confluent Cloud** (Kafka + Flink) on Azure. The platform processes payment transactions across 16 RBI-regulated payment channels (UPI, Cards, IMPS, NEFT, RTGS, Net/Mobile Banking), detecting fraud through a deterministic Confluent Flink core that fuses signals from multiple sources while maintaining complete explainability and regulatory compliance.
Core Innovation
The platform's architectural thesis is simple but powerful: **a deterministic Confluent Flink core does all detection and PII handling, so the regulated decision is explainable and replayable**. Because PII is HMAC-hashed at source and never leaves the deterministic layer in raw form, the hardest AI-governance problem — sensitive data reaching a model — is **eliminated upstream rather than policed downstream**. Flink is simultaneously the detection engine, the PII guardian, and the AI-governance enabler.
Platform Highlights
- **8 Confluent Flink SQL statements** implementing end-to-end fraud detection (no custom JAR operators)
- **3 managed Confluent Connectors** (PostgreSQL CDC source + 2 PostgreSQL sinks)
- **Confluent Tableflow** for zero-ETL analytical egress to Apache Iceberg lakehouse
- **33 Avro schemas** registered in Confluent Schema Registry with full compatibility governance
- **27 Kafka topics** following strict `<domain>.<entity>.<event>` naming convention
- **350ms P99 latency** for full ensemble decision (Flink rules +
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Checkpoint Current
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I'm working on it...
Auto-approval disabled
Cancel
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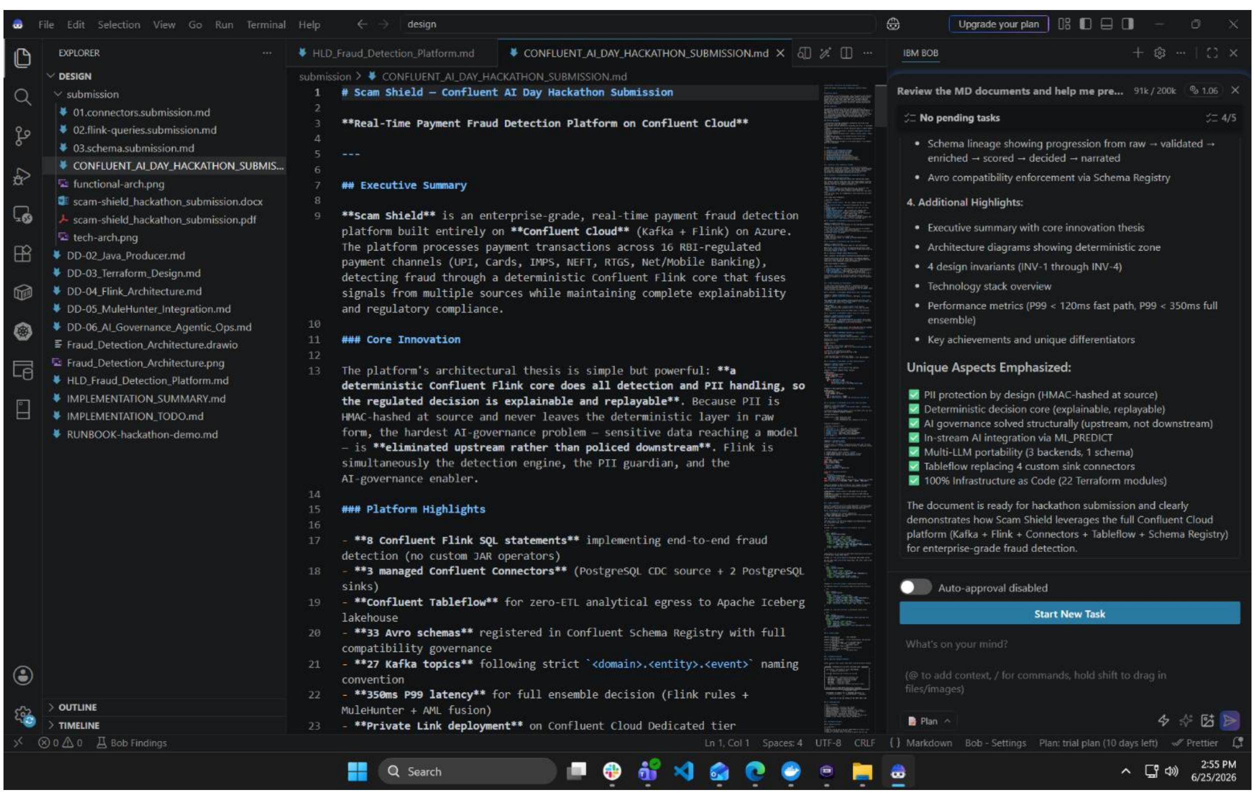

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- **Private Link deployment** on Confluent Cloud Dedicated tier
IBM BOB
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No pending tasks
4/5
Schema lineage showing progression from raw → validated → enriched → scored → decided → narrated
Avro compatibility enforcement via Schema Registry
4. Additional Highlights:
Executive summary with core innovation thesis
Architecture diagrams showing deterministic zone
4 design invariants (INV-1 through INV-4)
Technology stack overview
Performance metrics (P99 < 120ms fast path, P99 < 350ms full ensemble)
Key achievements and unique differentiators
Unique Aspects Emphasized:
PII protection by design (HMAC-hashed at source)
Deterministic decision core (explainable, replayable)
AI governance solved structurally (upstream, not downstream)
In-stream AI integration via ML_PREDICT
Multi-LLM portability (3 backends, 1 schema)
Tableflow replacing 4 custom sink connectors
100% Infrastructure as Code (22 Terraform modules)
The document is ready for hackathon submission and clearly demonstrates how Scam Shield leverages the full Confluent Cloud platform (Kafka + Flink + Connectors + Tableflow + Schema Registry) for enterprise-grade fraud detection.
Auto-approval disabled
Start New Task
What's on your mind?
(@ to add context, / for commands, hold shift to drag in files/images)
Plan
Bob Findings
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